

Information in Central Bank Sentiment: An analysis of Fed and ECB Communication*

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Abstract

This paper constructs a sentiment measure for published central bank (CB) communications using natural language processing and FinBERT techniques (machine learning models adapted to analyze financial news). We analyze sentiment of various Fed and ECB communications, but focus on FOMC minutes and ECB press conferences for our main results. We find that Fed and ECB sentiment move together. Using local projection analysis, we find that CB sentiment predicts policy interest rates and the Taylor rule several quarters into the future. Fed sentiment also leads ECB rates. In turn, stock market returns and deflation expectations tend to lead Central Bank sentiment.

Keywords: sentiment, textual analysis, central bank communication, Federal Reserve, European Central Bank, Taylor rule

JEL codes: C45, E50

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1 Introduction

In recent years, central banks have faced various crises, including the 2008 financial turmoil and the post-pandemic inflation surge in both the US and the Eurozone. In response, policymakers implemented measures to stabilize economies and markets. Communication has at times supported drastic policy actions. As emphasized by Blinder et al. (2024), such communication is crucial in guiding inflation expectations and signaling potential future policy moves. As they put it, “When central bankers talk, financial markets listen – intently”.

Central bank (CB) communication is usually defined as the provision of information to the public on the objective of monetary policy, the strategy to achieve these objectives, the economic outlook, and the outlook for future policy decisions (Blinder et al., 2008). CB communication has emerged as a cornerstone tool for shaping the monetary policy sentiments of all financial market participants. Its role has transformed from supporting the conventional monetary policy framework to actively guiding financial markets and helping central banks achieve their objectives (Nițoi et al., 2023).

Central banks communicate through various channels, including policy statements, post-meeting press conferences, economic forecasts, monetary policy reports, speeches, interviews, and testimony to parliament (Blinder et al., 2008). These communications are supposed to help reinforce key messages and provide additional clarity on the central bank’s policy intentions (de Haan and Hoogduin, 2024; Blinder et al., 2008; Benchimol et al., 2021).

Communication practices for both the Federal Reserve (Fed) and the European Central Bank (ECB) have evolved over the past several decades. Before the 1990s, central banks followed a practice of little communication that was indirect in its messaging (de Haan and Hoogduin, 2024). Over time, central bank communications and transparency have evolved dramatically and become more complicated. Although the tools and communication methods used by the Fed and the ECB were relatively different in the past, there is evidence that they have converged over time (de Haan and Hoogduin, 2024). For example, the FOMC began publishing minutes of its meetings in 1993, while the ECB started publishing its minutes in 2015. However, the ECB has held press conferences after each policy meeting since its

inception in 1999, while the Fed introduced press conferences after each meeting in 2019.

Although the communication tools and techniques of both the Fed and the ECB have become more similar, it remains an open question whether the link between communication and actual policy has done the same. Do the two banks react similarly to the state of the global economy? Are there feedback effects where the policy of one leads policy enacted by the other? For example, the ECB started lowering rates in June 2024, in light of a continued fall in inflation in the Eurozone. In contrast, the Fed at the time was expected to keep rates on hold. According to Reuters, ECB President Christine Lagarde said her institution was “data-dependent, not Fed-dependent.” However, analysts and policymakers remark that high US inflation and interest rates were bound to have an impact on the ECB’s plans via financial markets and trade.¹

In this paper, we explore the relationship between Fed and ECB monetary policy by constructing and analyzing the sentiment of Fed and ECB communications. While there is a large and growing literature aimed at quantifying sentiment from central banks’ announcements and speeches, we extend this analysis by developing a measure for sentiment over time for both the ECB and the Fed, comparing sentiment based on different communications, and analyzing Fed and ECB sentiment together.

The analysis begins by constructing a state-of-the-art measure of sentiment. We use the FinBERT pre-trained Natural Language Processing (NLP) model. FinBERT, an open-source model that has been specifically trained on financial data, outperforms almost all other NLP techniques for financial sentiment analysis.

We proceed in four steps. First, we construct sentiment for different central bank communications: minutes, press conferences, statements, the Beige Book, the Economic Bulletin, and speeches. The sample periods over which these are available are quite different. We apply an HP filter to the resulting time-series in order to filter out high-frequency noise and instead focus on lower-frequency movements in sentiment that reflect changes in economic

¹See: “ECB cannot ignore Fed as it goes down its rate cut path” by Francesco Canepa and Balazs Koranyi in: <https://www.reuters.com/markets/europe/solo-minded-ecb-may-find-itself-singing-feds-hymn-sheet-after-all-2024-04-11/>

conditions and the outlook for monetary policy.

The statements with the longest sample periods are Fed minutes and ECB press conferences. We choose these as our main measures of sentiment. This is partly because of the longer sample period but primarily because the other measures – beige book, economic bulletin, and speeches – tend to have low correlation with the main measures. This is to be expected for speeches, which are much more frequent and often are not as focused on the overall outlook for policy. Fed statements, in contrast, have a fairly high and quite stable correlation with minutes, supporting our choice of using minutes as our main measure.

Second, using local projection analysis, we ask if the Fed or the ECB leads the level of sentiment at the other central bank. We find some contemporaneous correlation, i.e. that sentiment tends to move together, but there is little evidence of policy predictability. That is, there is little evidence that policy at one central bank leads the other. There is some limited evidence of low ECB sentiment leading higher Fed sentiment several quarters in the future.

Third, we explore the dynamic relationship between Fed and ECB sentiment, the state of the economy, and the monetary policy stance. We use local projection analysis to assess whether or not sentiment reflected in CB communications leads the Taylor rule and policy interest rates at the Fed and the ECB. We find that for both the Fed and the ECB, sentiment leads economic conditions as measured by the Taylor rule, i.e. the output gap and the deviation of inflation from its target. Similarly, sentiment leads policy interest rates for both central banks. The effect is economically large and statistically significant. It builds up over the first three quarters and is present several quarters into the future. One potential concern is that central bank sentiment may simply be a measure of overall consumer sentiment. However, we find that Fed sentiment contains additional information. Indeed, Fed sentiment is a more informative predictor of the federal funds rate as compared to the Michigan sentiment measure.

We also find that Fed sentiment leads ECB policy rates, while ECB sentiment has a much weaker effect on the Fed funds rate. In addition, the exchange rate is also affected. Higher

Fed sentiment leads US dollar appreciation relative to the Euro.

Fourth, we explore drivers of CB sentiment. Financial markets generally react very quickly to new information (Malkiel, 2003), suggesting that sentiment may reflect public information about future prospects, which in turn is reflected in stock returns. That is, if one hypothesizes that financial markets reflect all publicly available information, stock returns should lead sentiment. We find evidence supporting this hypothesis for the Fed – higher stock returns lead higher sentiment several quarters into the future. For the ECB the effect is much weaker. We also find that a measure of deflation (Hilscher et al., 2024) tends to lead lower sentiment both for the Fed and the ECB.

The remainder of the paper is organized as follows. Section 1.1 outlines the methods used to construct sentiment measures, specifically FinBERT, while section 1.2. summarizes the related literature. Section 2 discusses our sentiment measure, and the Fed and ECB communication data that we analyze; we also compare sentiment across measures, while focusing on our two main measures, minutes (Fed) and press conference (ECB) sentiment. Section 3 uses local projection analysis to estimate the dynamic effect of sentiment on the Taylor rule rate, interest rates, and the exchange rate. Section 4 explores potential drivers of sentiment such as stock market returns and deflation probabilities, Section 5 discusses data limitations when constructing CB sentiment, and Section 6 concludes.

1.1 Methods for Sentiment Analysis

Analyzing central bank communications and annotating the sentiment of every published sentence in various documents is a time-consuming task. In the past, this has restricted sentiment analysis projects to, e.g., only a narrow range of publications or a restricted sample period (Lucca and Trebbi, 2009; Wilson et al., 2005). As computing technology has improved, potential applications in various areas, including financial market analysis, have also advanced. Initially, many papers translated textual information from central bank communications into quantitative measures through the use of lexicon-based approaches. This method relies on predefined lists of words, known as lexicons or dictionaries, where each

word is assigned a score indicating its sentiment. For example, a lexicon might categorize words as positive, neutral, or negative, with corresponding scores of 1, 0, and -1, respectively (Boschen and Mills, 1995).

Word-matching methods are often referred to as bag-of-words (BOW) techniques.² This is because they disregard contextual characteristics such as word order, part-of-speech, and co-occurrence with other words, focusing solely on the individual words themselves. Shapiro et al. (2022) and Loughran and McDonald (2011) note that general dictionaries do not provide sufficient accuracy for assessing tonality in finance contexts and have created a dictionary tailored to the context of 10-K reports. They find that almost three-fourths of the words in the Harvard dictionary have different connotations in finance.

The main limitation of lexicon-based methods is their inability to account for the context of keywords (Nițoi et al., 2023). Therefore, a second generation of Machine Learning (ML) methods has been developed. These methods construct complex models and use probabilistic forecasting to find the sentiment of a given set of text. This method relies on annotated training data (e.g., sentences labeled as positive, negative, or neutral) to learn the relationships between text features and sentiment labels. Once trained, the model can then predict the sentiment of new, unseen text (Bochkay et al., 2023). However, this type of model ignores the order of words, and the small training sets can pose modeling challenges (Shapiro et al., 2022).

The third generation of models are the deep learning models that extract sentiments from a text by relying on similar labeled text data to learn and tune its embedding parameters. The main shortfall of these models is the need for a large amount of human-labeled text data for training (Kaplan et al., 2020; Hoffmann et al., 2022; Wilson et al., 2005).

In this paper, we use a third-generation model, FinBERT, for sentiment analysis. Bidirectional Encoder Representations from Transformers, or BERT, is a transformer deep learning model developed by Google in 2017 and was made open source in 2018. The significant

²Although it is rare, lexicons can be a list of n-grams (several words) rather than just unigrams (single words). Similarly, some lexicons use word stems to match all inflected variants of a given word.(Shapiro et al., 2022)

advancement that BERT made in the field of machine learning is its ability to be trained on large datasets. This is because BERT does not require much preprocessing as previous models such as Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) did. Bert is the implementation of Google to the type of models which are called “attention-based model”, or transformer. It operates by adjusting its focus, highlighting the most relevant and important context clues while downplaying less significant information—a technique referred to as “soft attention” (Vaswani, 2017).

Building on top of the BERT model, FinBERT was first developed by Araci (2019) and is a transformer model designed to analyze financial documents. Since FinBERT was trained on financial sentiment lexicons, it can predict the sentiment of financial documents with a higher accuracy than BERT and other similar models. Further improvements have been made to both the model and preprocessing of the data used in the model. By training the model on central bank communications, (Gössli et al., 2023) were able to better identify the sentiment of announcements. It is this version that we use in our analysis.

1.2 Related Literature

There are two main strands of related literature. The first focuses on central bank communication sentiment analysis to understand how CB sentiment affects market behavior and vice versa. Initially, this literature used keyword lexicons, categorizing words as positive or negative, or hawkish or dovish. Documents were analyzed by counting the frequency of different words, considering their context, such as the proximity of “raise” or “lower” to “interest rate.” This method was pioneered by de Haan et al. (2007) who study how ECB communications influenced Eurozone inflation expectations. This ML-based approach improves on previous methods by increasing both the volume of texts that can be analyzed and enabling the model to better understand the nuances of financial publications.

An important part of lexicon-based keyword analysis is the financial dictionary created by Loughran and McDonald (2011) that uses 10-K reports to measure sentiment of financial documents. However, the interpretation of words as negative or positive may differ in the

context of central bank communication. To address these issues, Correa et al. (2021) refined and improved the dictionary. Still, the drawbacks of the keyword approach remain - there is still a lack of context and nuance.

To enhance automated sentiment analysis and improve understanding of context and nuance, FinBERT was developed by building on Google’s BERT model (Araci, 2019; Devlin et al., 2019). BERT considers the context of each word by reading both to the left and right within the text. This capability is further enhanced for financial texts by adding an additional level of training and fine-tuning, which was done to create the FinBERT model (Araci, 2019). This model has been used to increase the accuracy of sentiment analysis for both general financial texts and specific areas of finance, including central bank publications (Gössi et al., 2023; Yang et al., 2020; Gu et al., 2024; Wang, 2023). However, these previous analyses typically focused on one set of texts from a single central bank at a time. The high cost of sentiment analysis when performed manually, or the high computing power previously needed when using automated analysis, has hindered the comparison of multiple publications across a large time span (Lucca and Trebbi, 2009; Wilson et al., 2005).

These measures of sentiment have also been used to analyze stock market performance, trade volume, and key macroeconomic variables including interest rates and the Taylor Rule (Picault and Renault, 2017; Gu et al., 2022; Jubinski and Tomljanovich, 2013; Tadler, 2022). The literature also assesses the effect of sentiment on market stability (Correa et al., 2017). Correa et al. (2021) found that the sentiment of central bank communications is a strong indicator of the occurrence of a banking crisis. They developed a financial stability sentiment (FSS) index, which measures the negative sentiment of a country’s central bank communications. They found that a one percent increase in their FFS index led to a twenty-nine percent increase in the probability of a banking crisis.

Furthermore, Oosterloo et al. (2007) analyze not only how the information within the publications affects the market, but also how central bank communication methods and transparency have changed their ability to stabilize the market and affect public perception. This is especially relevant over the past two decades in the US and EU as CB communication

methods and transparency have converged. In 2015 for example, the ECB overhauled its communication strategy publication style and schedule more closely with that of the Fed (Gardt et al., 2022).

The second strand of related literature studies the relationship between the ECB and Federal Reserve monetary policy. Belke and Gros (2005) find that the ECB is often influenced by the Fed, but the reverse is true at least as often when considering longer sample periods. (Ehrmann and Fratzscher, 2005) find that spillover effects are stronger from the US to the Eurozone. Chinn and Frankel (2007) argue that nominal US interest rates influence European rates unilaterally, even after the establishment of the European Monetary Union (EMU). We find some evidence that Fed US CB sentiment leads ECB sentiment, though the reverse is also true to some extent.

The synchronized behavior of central banks is also measured by deviations from the Taylor rule. Hofmann and Bogdanova (2012) argue that deviations from the Taylor rule can best be interpreted as a change in the global equilibrium real interest rate. Taylor (1993) argues that a further transmission channel for international spillovers stems from the fact that central banks no longer set policy rates independently. Our focus is sentiment, and indeed we find that sentiment tends to move together.

2 Sentiment of Central Bank Communications

2.1 Data on CB Communications

Both the Federal Reserve and the European Central Bank (ECB) regularly publish various documents that provide valuable information about monetary policy and economic conditions. For this analysis, we focus on the Federal Reserve’s Minutes, Statements, Beige Books, and Speeches. In the case of the ECB, we examine Press Conferences, Monetary Policy Accounts, Economic Bulletins, and Speeches.

The FOMC holds eight regularly scheduled meetings each year, with additional meetings convened as necessary to discuss the U.S. economic outlook. After each Federal Open Market

Committee (FOMC) meeting, the Fed issues a policy statement summarizing the Fed’s policy decisions and forward guidance, such as changes to interest rates or asset purchases. FOMC decisions have a significant impact on the economy. For example, the FOMC’s decision to raise or lower interest rates can have a major impact on borrowing costs, investment, and economic growth.

The Federal Reserve began holding regular post-FOMC press conferences in 2011 under then-Chair Ben Bernanke. Initially, press conferences occurred only after every second meeting (following the quarterly meetings when the Fed released its economic projections). However, only in 2019, the frequency increased to after every FOMC meeting. Due to the relatively short history of press conferences we cannot based our analysis on this type of communication for the Fed.

About three weeks after each meeting, the minutes are published. The FOMC meeting minutes provide a more detailed account than the post-meeting statement. The minutes of FOMC meetings are our primary source for measuring Fed sentiment, as they provide the most detailed account of the discussions and considerations that lead to policy decisions, and they have the longest history dating back to 2000. They offer transparency into the central bank’s decision-making process and help investors and economists understand the rationale behind policy changes. For example, the minutes might reveal that the Fed was divided on the decision to raise rates, with some members expressing concerns about the potential impact on economic growth.

Another type of communication that we analyze is the Beige Book, published eight times a year. It summarizes economic conditions in each Federal Reserve district, providing both qualitative and quantitative information on employment, production, spending, and credit conditions. This contributes to a comprehensive understanding of regional economic activity. For example, a Beige Book might report that manufacturing activity in the Midwest is expanding, while consumer spending in the Northeast is slowing down.

Finally, we refer to speeches by Federal Reserve officials, such as the Chair or Governors. These speeches provide insights into their views on the economy, monetary policy, and fi-

nancial markets. They can signal potential policy changes or updates on the central bank’s assessment of the economic outlook. For example, a speech by the Fed Chair expressing concern about rising inflation could indicate a possible interest rate hike in the future. Recently, Powell’s speech at the 2024 Jackson Hole conference was viewed as a signal of upcoming rate decreases at the September FOMC meeting. However, unlike the Minutes, Statements, and Beige Books, speeches by Fed governors are not released on a routine schedule but are instead determined by their decision to respond to economic conditions and the timing of events to which they are invited.

The European Central Bank also has a number of publications that provide insights into its assessment of the economic situation and its policy decisions. Similar to Fed’s Statements, Monetary Policy Accounts summarize the discussions and decisions of the ECB’s Governing Council and released right after each meeting. Meeting accounts (similar to FOMC minutes) are released four weeks after the meeting. These offer insights into the deliberations of the Governing Council, including dissenting opinions. Similar to the Fed, the ECB held press These are held after Governing Council meetings offer ECB officials an opportunity to explain the central bank’s policy decisions and provide insights into the ECB’s thinking and its assessment of the economic situation.

The two additional resources that we refer in our analysis are the Economic Bulletin, published quarterly, provides an in-depth analysis of economic and monetary developments in the euro area, and speeches by ECB officials, such as the President or Executive Board member. As for the Fed, these speeches often signal potential policy changes or provide updates on the central bank’s assessment of the economic outlook.

Table 1 provides summary statistics for the various documents that we analyze. The sample period for more than half of the documents, including our main measures, begins in 2000. The most frequent form of communication is speeches, as they are not bound to a fixed meeting schedule or specific release dates. In our analysis of central banks’ sentiment following monetary council meetings, we primarily focus on the Minutes for the U.S., as we have data from 2000 and these documents provide much more detail than the Statements.

For the EU, we similarly focus on the press conferences for the same reasons.

Table 1: Central Bank Publications for Sentiment Analysis.

The number of documents per year is based on publication standards as of December 2023 and is averaged across sampled publications.

Dataset	Start Date	End Date	Num. of Docs.	Docs/Year	Words Per Doc
Fed Documents					
Minutes	02-2000	05-2023	188	8	13,692
Statements	02-2000	05-2023	199	8	475
Beige Books	01-2006	05-2023	140	8	20,435
Speeches	01-2000	05-2023	1,225	52	3,698
ECB Documents					
Press Conferences	01-2000	05-2023	241	8	1,634
Monetary Policy Accounts	02-2015	06-2023	69	8	5,704
Economic Bulletin	02-2015	05-2023	67	8	61,710
Speeches	01-2000	06-2023	1,788	76	4,502

Over time, the communication and transparency of the FOMC and the ECB have changed. Middeldorp (2011) claims that the FOMC has become more transparent and has increased communication to enhance monetary policy predictability. Middeldorp (2011) identifies FOMC communication reforms in the early 1990s and in 2003 that significantly improved monetary policy predictability.

2.2 Constructing Sentiment using FinBERT

To obtain an accurate sentiment classification of publications, we process, classify, and filter these various texts. While this may be automated in the future, we are currently unable to feed an entire raw document into a machine learning model and receive an accurate result. Instead, we process the data by classifying each sentence. Once we have results for every sentence in each document, we group and filter these results to further improve the accuracy of our sentiment measure.

To measure sentiment, we implement a modified version of FinBERT, a pre-trained NLP model based on Google’s BERT architecture. The original FinBERT model is trained on financial documents as 10-K filings, but not on monetary policy documents as FOMC Minutes. We use the version by Gössi et al. (2023), which has been specifically trained on FOMC minute publications. Further, this version is more suitable for our purposes, as our goal is to evaluate the overall sentiment of each sentence, rather than to classify sentences by specific characteristics, as done by Tadler (2022), who labels them as hawkish or dovish.

In our analysis, we define a positive sentiment as one that is favorable in a general financial context, not limited to a specific economic factor. This broader sentiment classification allows us to address various questions related to central bank sentiment, without focusing exclusively on particular issues, such as inflation.

After classifying the sentiment of our processed texts, we apply the Hodrick-Prescott filter, which removes the cyclical component of a time series from raw data. This filter provides a smoothed-curve representation of a time series, making it more sensitive to long-term trends as opposed to short-term fluctuations.

2.2.1 Data Processing and Sentiment Classification

For the natural language processing (NLP) filtering of the data, the Python library Natural Language Toolkit (Bird et al., 2009) is leveraged to read and break up each publication. The texts are broken up into the detected sentence sub-strings using the sentence tokenizer API. Once the text is broken up into sentences, the texts are further refined in preparation for sentiment analysis. We accomplish this by removing punctuation symbols and stop words. This makes it easier for FinBERT to analyze the essence of the sentence and have a higher chance of accurately identifying it. Sentences are identified using the NLTK toolkit sentence tokenizer API (Bird et al., 2009).

We then pre-process the text by using the technique described in (Gössi et al., 2023). Unnecessary commas are removed, condensing sentences to their essential components, while maintaining the overall sentiment of the sentence. Refining the sentence is achieved by ad-

justing the focus of sentences based on specific conjunctions like “but,” “although,” and “though,” as well as the position of commas. This function isolates the most sentimentally relevant part of a sentence, especially in complex sentences with contrasting clauses ³ The resulting structure is cleaner and more coherent, which can improve sentiment analysis accuracy.

Each sentence is processed using the FinBERT model and assigned a score of -1, 0 or 1. The individual sentence-level scores are averaged in the document to arrive at the final sentiment of the communication. Table 2 presents summary statistics for all of the documents that we analyze. Overall, sentiment is slightly negative on average over the sample period for the Fed, while averages are split between positive and negative for the ECB. However, more important for our analysis is that there is substantial variation in sentiment. Minutes and Press Conferences – our main two measures – both reflect variation over time; sentiment is slightly negative for the US, while it is slightly positive for the Eurozone.

³For example, in sentences beginning with “but,” the text following “but” is isolated as it likely contains the crux of the sentiment. Similarly, for sentences containing “although” or “though,” the function extracts the most sentimentally significant portion by navigating the commas that typically demarcate shifts in sentiment.

Table 2: Summary Statistics of Central Bank Sentiment

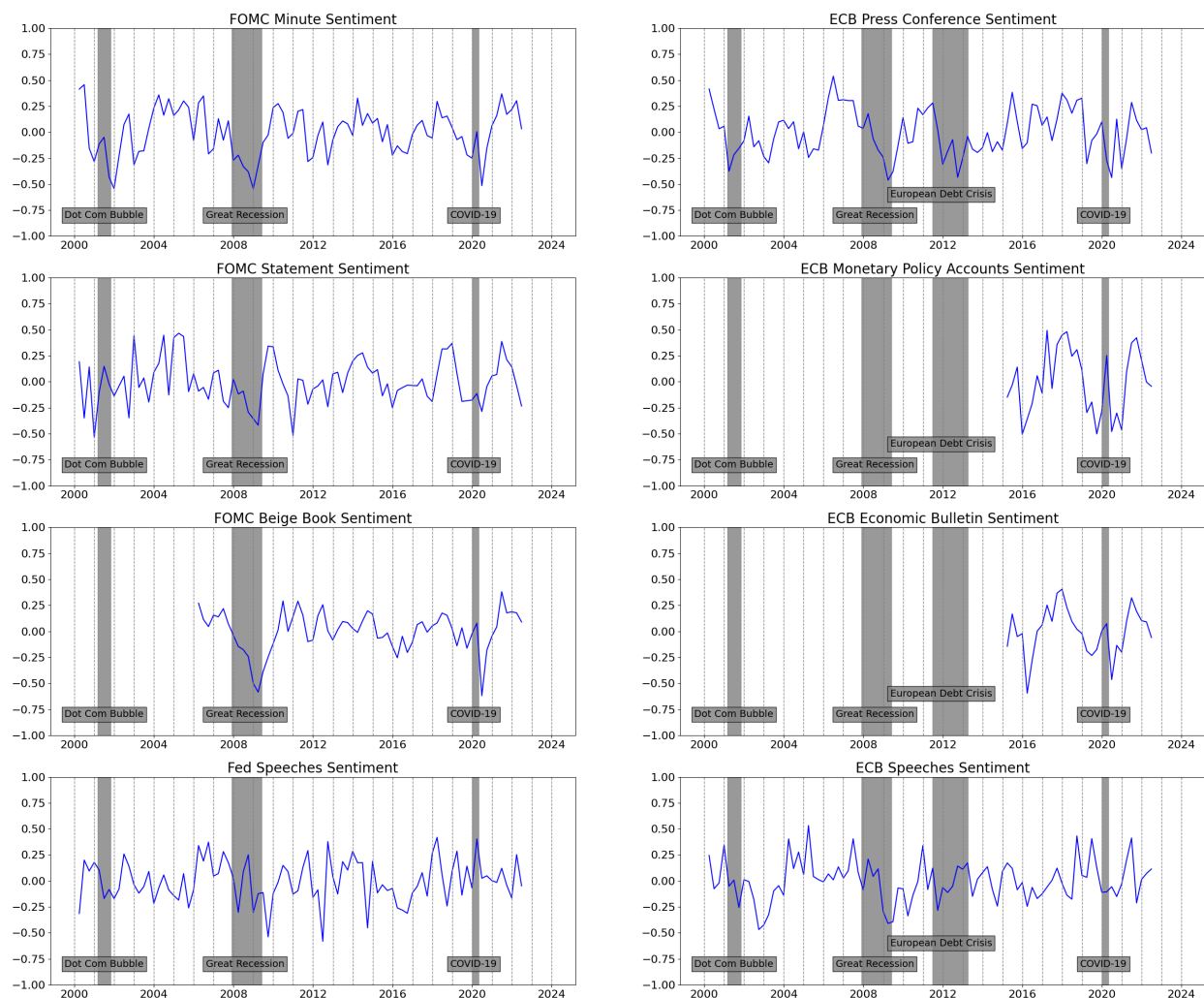
The sentiment statistics for each publication are calculated by averaging the sentiment of all sentences across the publications in our sample.

Dataset	Mean	SD	Min	Max	Num. of Docs.
Fed Documents					
Minutes	-0.14	0.11	-0.42	0.16	188
Statements	-0.04	0.18	-0.83	0.31	199
Beige Books	-0.04	0.14	-0.46	0.18	140
Speeches	-0.03	0.14	-0.40	0.53	1,225
ECB Documents					
Press Conferences	0.07	0.16	-0.32	0.45	241
Monetary Policy Accounts	-0.12	0.12	-0.36	0.11	69
Economic Bulletin	0.03	0.04	-0.10	0.10	67
Speeches	-0.01	0.11	-0.40	0.43	1,788

2.2.2 Lower frequency sentiment: HP Filter

Communications happen at different frequencies and at different times. In order to standardize the data, as a first step, we average sentiment scores for each communication and quarter. However, CB communications, while structured and consistent, are subject to inherent volatility arising from economic events, policy announcements, and, as highlighted in our study, changes in communication policies themselves. The raw sentiment score time series will reflect variation at different frequencies. For our analysis, we want to isolate lower-frequency movements and therefore apply the Hodrick and Prescott (1997) (HP) filter to the quarterly data. By eliminating noise induced by temporary events, we ensure that variation of the filtered sentiment series accurately reflects the message of Central Bank communications over time (Fernandez et al., 2021).

Figure 1: Central Bank Sentiment Over Time. Sentiment measurements for Fed and ECB publications, quarterly, HP filtered. Important contractionary periods shaded (NBER recessions for the US, OECD-based recession indicator for the Eurozone).



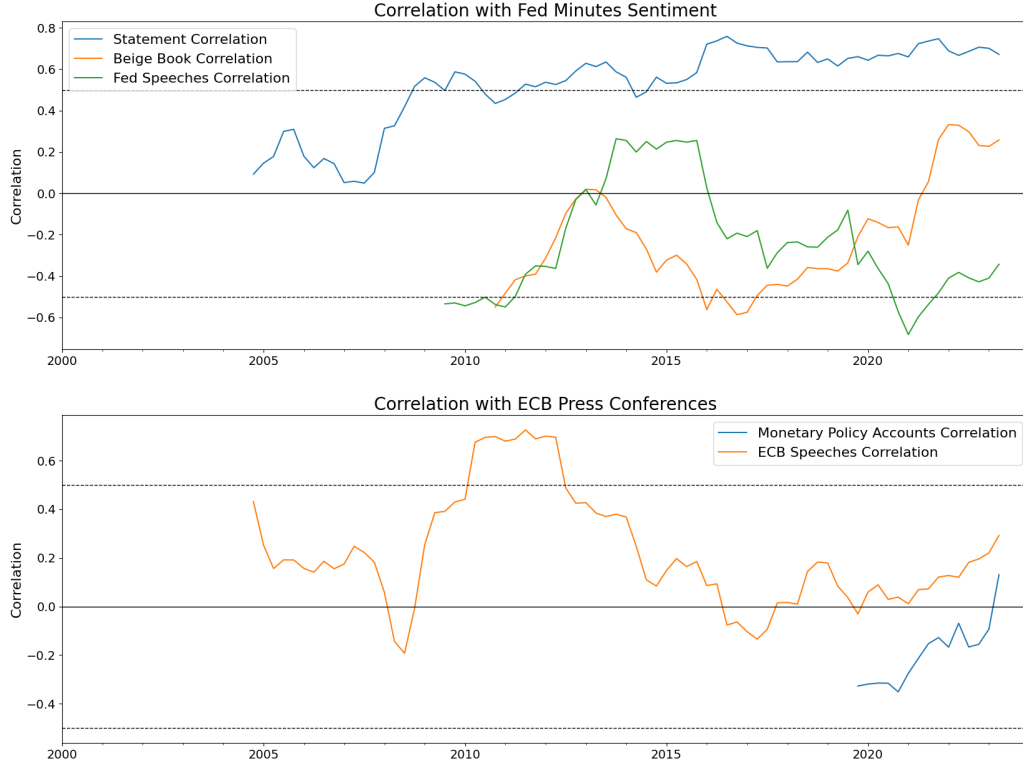
To help interpret the time series patterns in the different measures, we add shaded areas for important contractionary events, specifically the aftermath of the Dot-com bubble bursting, the Great Recession, the European Debt Crisis, and the Global Pandemic. Sentiment tends to move with the cycle, declining during contractions and is generally higher and often increasing during expansions.

2.2.3 Comparison across publications

We next compare sentiment levels across various publications, both within a Central Bank and between the Fed and the ECB. We choose FOMC minutes and ECB press conference sentiment as the respective benchmarks. Both are shown at the top of Figure 1. FOMC minutes are chosen over FOMC statements because they contain a more detailed description of FOMC views. They are more detailed and longer when compared to FOMC statements due to being minutes for their entire meeting as compared to the relatively shorter statement given to the public (Tadle, 2022; Apergis, 2015). ECB Press Conferences are used as the representative text for The ECB because it represents the general sentiment of The ECB and its publication covered the entire time frame of our analysis. While ECB Economic Bulletins would be a closer mirror of FOMC minutes, given this text set only started being published in 2015, the sample is not sufficient to cover the entire range of our analysis (Kaminskas and Jurkšas, 2024).

We find that different measures are not always in step with each other, though this is not always the case. Five-year rolling correlations between our primary sentiment measure (minutes and press conferences) and the other measures are shown in Figure 2. For the ECB, only speeches are available farther back. Correlations with speech sentiment are variable and somewhat unstable, both for the Fed and for the ECB, though the correlation is lower for the US. This pattern reflects the potential differences in messaging that Central Banks have for different communication methods. Fed minutes and ECB press conferences are scheduled, planned events that each party knows will be analyzed in depth by the public. In contrast, speeches have a higher frequency and may be delivered by a range of speakers; the sentiment of these transcripts is not going to be as carefully planned and will differ across speaker and topic. The correlation with Fed beige book sentiment is similarly variable and tends to be low or negative. In contrast, the correlation with Fed statements is generally much higher and more stable.

Figure 2: Correlation of Sentiment Measurements. Five year window correlation of Central Bank publications relative to reference publication sentiment (Fed minutes and ECB press conferences). Due to the shorter sample period, correlations for Economic Bulletin sentiment is not included.



Within the set of Fed publications, we observe a steady growth in the correlation between minute sentiment and statement sentiment. This shows a growing alignment between the sentiment presented in their meeting statement and their conversations during meetings. This is in contrast to beige books and speeches, which have a similar correlation to minutes but fluctuate more than statement correlation and do not have the same trend towards consistency over time.

For the ECB, due to publication history, we focus on the relationship between press conferences and speeches. Between these two publications, the correlation in sentiment is variable in the early 2000s and starts declining in 2012/13. After that, we observe a

generally lower correlation, hovering around zero. This pattern confirms that speeches are less appropriate for sentiment measurement.

3 Effects of Sentiment on Macro Variables

For our main analysis, we focus on FOMC Minutes and ECB Press Conferences. FOMC minutes cover a long time span and have been used in previous sentiment analysis. For the ECB, we correspondingly use press conferences since these are the natural comparison to FOMC minutes. It is also the only time series, apart from speeches, with a sufficiently long sample period; Monetary Policy Accounts only start in 2015.

Our goal is to analyze the dynamic relationship between sentiment and important macroeconomic variables. We use a local projection framework to explore this question. Specifically, we estimate whether or not sentiment, which we hypothesize summarizes the information in Central Bank communications, contains information about sentiment at the other Central Bank, future Fed and ECB policy rates, Taylor rule rate, and the EUR/USD exchange rate. To account for any spurious effects in the regression, we include the lagged dependent variable as a control (Granger and Newbold, 1974). Following Montiel Olea and Plagborg-Møller (2021); Gafarov et al. (2024) we report White standard errors to correct for homoscedasticity within our data (White, 1980).

Local projection analysis is a statistical technique that can be used to analyze the temporal relationship between two variables (Jorda, 2005). We apply this in our analysis to assess the magnitude of which Central Bank sentiment affects interest rates and the Taylor rule. Next, we identify determinants of sentiment by using stock returns and inflation expectations as a leading indicator.

3.1 Federal Reserve and European Central Bank sentiment

Before proceeding with the analysis of sentiment and macro variables, we first ask if one central bank’s sentiment tends to lead the other. Given that central banks have taken steps

to align their publication and meeting strategies, it is possible that sentiment may also be somewhat aligned.

Specifically, we use local projection analysis to ask, for example, if ECB sentiment at time t affects Fed sentiment at time $t + h$. Table 3 reports the results (Panel A for the effect of ECB sentiment on Fed sentiment and vice versa in Panel B). We find that Fed and ECB sentiment move together. There is a large, statistically and economically significant effect of ECB sentiment on Fed sentiment and vice versa. These effects are present after controlling for lagged sentiment.

These results demonstrate the similarity in policy outlook across these two major currency areas. For instance, an optimistic sentiment in the ECB's communications about the Eurozone's economic recovery might come at a similar time as a positive outlook by the Fed's communications, possibly due to improved conditions in the US and the Eurozone positively impacting the global and the other economy.

This result could also imply that there is some degree of synchronization in the policy outlooks of the two Central Banks, possibly due to shared economic challenges or global economic conditions that affect both the Eurozone and the United States. In a highly interconnected global economy, coordinated Central Bank policies can help mitigate international financial turbulence. Market participants can use this information to anticipate shifts in monetary policy and adjust their investment strategies accordingly.

Our next finding is that ECB sentiment leads Fed sentiment next period (Table 3 column 2). However, looking at future quarters, the sign of the effect switches and there is a negative effect of ECB sentiment on future Fed sentiment, three, four, and five quarters in the future. We also notice some autocorrelation. The interpretation is that ECB sentiment tends to improve ahead of Fed sentiment, but only temporarily. At longer horizons, there is a negative effect.

For the ECB, there is less autocorrelation. There is also no evidence that Fed sentiment leads ECB sentiment. Our results thus show that sentiment in one bank's communications can lead the sentiment in the other, but that there is little evidence of a strong relationship.

The one exception is a somewhat pronounced medium-term negative effect of ECB sentiment on future Fed sentiment.

Table 3: Dynamic Effect of Fed and ECB sentiment on the other Central Bank.

Panel A: Dynamic effect of ECB sentiment on Fed sentiment					
	Fed(t)	Fed(t+1)	Fed(t+2)	Fed(t+3)	Fed(t+4)
ECB(t)	0.184** (0.081)	0.186** (0.086)	0.097 (0.090)	0.021 (0.092)	-0.157** (0.074)
ECB(t-1)	-0.056 (0.077)	-0.161* (0.090)	-0.236** (0.097)	-0.329*** (0.087)	-0.203** (0.082)
Fed(t-1)	0.478*** (0.087)	0.219** (0.096)	0.207** (0.101)	0.242** (0.101)	0.067 (0.099)
Observations	89	89	89	89	89
R^2	0.310	0.095	0.085	0.170	0.156
Panel B: Dynamic effect of Fed sentiment on ECB sentiment					
	ECB(t)	ECB(t+1)	ECB(t+2)	ECB(t+3)	ECB(t+4)
Fed(t)	0.294** (0.116)	0.160 (0.161)	0.230 (0.155)	0.198 (0.149)	0.169 (0.155)
Fed(t-1)	-0.060 (0.142)	0.139 (0.154)	0.066 (0.164)	0.094 (0.163)	0.190 (0.153)
ECB(t-1)	0.481*** (0.082)	0.163 (0.102)	0.112 (0.101)	-0.055 (0.114)	-0.166 (0.113)
Observations	89	89	89	89	89
R^2	0.315	0.092	0.072	0.039	0.065

Note: The table shows results from regressing Fed sentiment at $t + h$ on ECB sentiment at t and $t - 1$ (Panel A); we control for lagged Fed sentiment. Similarly, Panel B shows results when regressing ECB sentiment on Fed sentiment. Following Montiel Olea and Plagborg-Møller (2021) we use White Standard Errors (in parentheses); * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

3.2 Effect of Sentiment on Interest Rates and Exchange Rates

We next consider the relationship between economic fundamentals and sentiment. Specifically, we ask if sentiment is a leading indicator for the Taylor rule and policy rates.

3.2.1 Effect of Sentiment on Taylor Rule rate

The Taylor rule captures the interest rate relationship with inflation and the output gap following (Taylor, 1993): Specifically:

$$r = \pi + 0.5y + 0.5(\pi - 2) + 2 \quad (1)$$

where

r is interest rate

π is rate of inflation over the previous four quarters

y is percent deviation of real GDP from a target

$$y = \frac{100(Y - Y^*)}{Y^*}$$

Y is real GDP

Y^* is trend real GDP

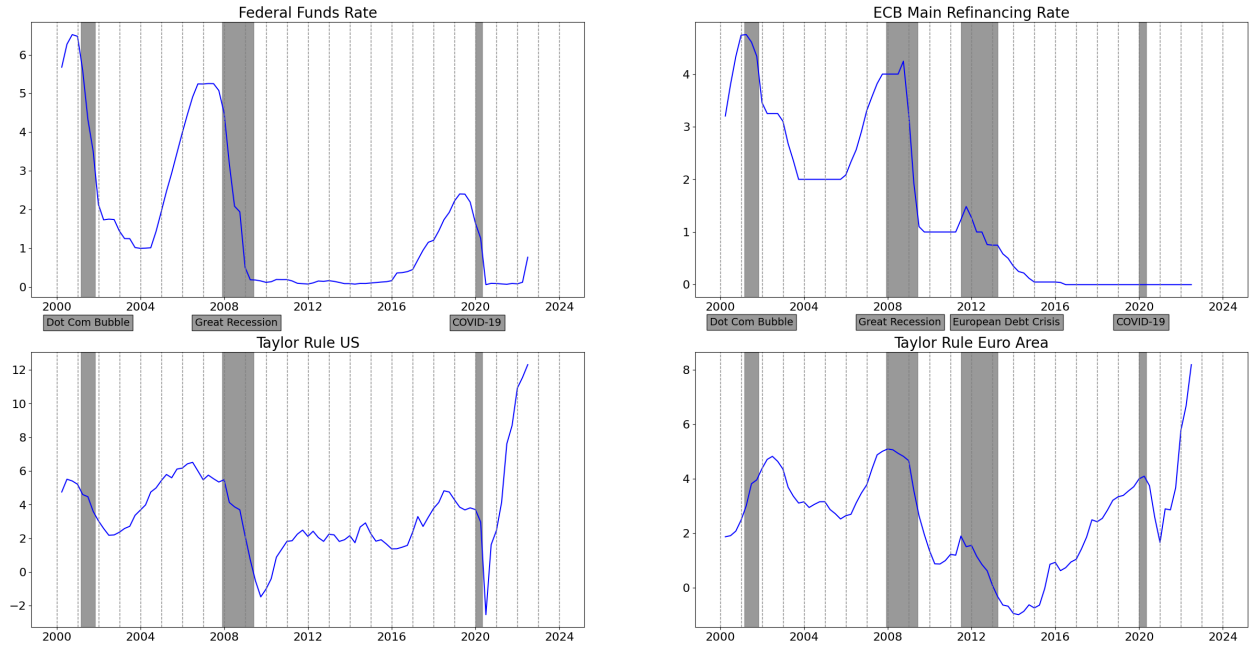
Figure 3 shows the Taylor rule interest rate for the US and the Eurozone. We plot this along with the main actual policy rates. Reflecting changes in inflation and output, we note that the Taylor rule rate drops during times of contraction and increases during expansions. A very striking pattern is the sharp increase in the Taylor rule rate toward the sample, in line with the sharp post-Pandemic spike in inflation. However, in contrast to the rest of the sample, actual policy rates do not move and become disconnected from the Taylor rule.⁴

Given that the Taylor Rule rate, policy interest rates, and sentiment all reflect economic

⁴For additional discussion of recent changes in inflation expectations and monetary policy see Hilscher et al. (2024).

fundamentals, we next ask whether or not sentiment leads the time series in Figure 3. Our hypothesis is that sentiment may lead both interest rates. As economic conditions improve and sentiment increases, the Taylor rule interest rate will increase, either because inflation picks up or because output growth is outpacing the trend. CB sentiment may reflect expectations of future changes in output and inflation and, thereby, lead the series.

Figure 3: Interest Rate and Taylor Rule for Fed and ECB Interest rates and Taylor Rule are given as quarterly statistics due to the publication frequency Taylor Rule input variables.



We start by considering the dynamic relationship between sentiment and the Taylor Rule. Table 4 Panel A reports results for the US; Panel B for the Euroarea. The results indicate a significant relationship between the sentiment of Fed and ECB communications and the Taylor-rule rate over the next several quarters. Starting at time t , positive sentiment in the communications is associated with a contemporaneous increase in the Taylor Rule rate, suggesting that optimistic central bank sentiment can signal a tightening monetary policy in response to expected inflation or economic growth. Conversely, negative sentiment might indicate a more accommodative monetary policy stance. Further into the future, the effect is stronger, so that positive sentiment today affects Taylor rule rates one, two, three and four quarters in the future.

These findings are significant because they highlight the predictive power of central bank sentiment on macro fundamentals. They suggest that central bank communications carry important forward-looking information, i.e. that they reflect not only current economic conditions, but also future anticipated developments including policy. These results demonstrate the importance of textual analysis and sentiment measurement in understanding central bank communication and monetary policy.

Table 4: Dynamic Effects of Minute and Press Conference Sentiment on Taylor Rule Rate

Panel A: Dynamic effect of Fed sentiment on Taylor Rule rate					
	Taylor US(t)	Taylor US(t+1)	Taylor US(t+2)	Taylor US(t+3)	Taylor US(t+4)
Fed(t)	6.033*** (2.310)	7.206*** (1.651)	10.656*** (2.278)	12.444*** (3.078)	11.736*** (4.088)
Fed(t-1)	-0.073 (1.767)	3.078 (2.284)	3.172 (2.408)	2.277 (3.105)	1.969 (3.552)
Taylor US(t-1)	0.957*** (0.055)	0.865*** (0.074)	0.733*** (0.092)	0.560*** (0.116)	0.373*** (0.122)
Observations	89	89	89	89	89
R^2	0.865	0.741	0.607	0.433	0.268
Panel B: Dynamic Effect of ECB sentiment on Taylor Rule rate					
	Taylor EA(t)	Taylor EA(t+1)	Taylor EA(t+2)	Taylor EA(t+3)	Taylor EA(t+4)
ECB(t)	1.019* (0.570)	1.962** (0.927)	4.280*** (1.252)	4.855*** (1.627)	4.794** (2.126)
ECB(t-1)	1.367** (0.681)	3.123*** (1.055)	2.760** (1.258)	2.925 (1.783)	3.498 (2.150)
Taylor EA(t-1)	1.018*** (0.030)	1.003*** (0.055)	0.989*** (0.084)	0.945*** (0.110)	0.889*** (0.129)
Observations	89	89	89	89	89
R^2	0.939	0.846	0.725	0.585	0.466

Note: The table reports results from regressing Taylor rule rates (Fed, Panel A, and ECB, Panel B) at $t + h$ on sentiment at t and $t - 1$; we control for lagged Taylor rule rate. We report White Standard Errors (in parentheses); *p<0.1; **p<0.05; ***p<0.01.

3.2.2 Effect of Sentiment on Interest Rates

A key role that Central Banks play in policy making is setting interest rates. Speeches and publications can include explicit indicators regarding future changes in monetary policy (Shapiro and Wilson, 2021), but less obvious indicators like sentiment and tone – variation that is captured by our measure – can also have an effect on the market (Gu et al., 2022). To analyze this connection, we explore the dynamic relationship between interest rates set by the Central Bank and sentiment.

Table 5 reports the results of local projection regressions of sentiment at time t on interest rates at time $t+h$. We find that, in both Europe and the US, sentiment has a strong positive relationship with the respective interest rate (Nițoi et al., 2023). Similar to the Taylor rule rate, the effect builds up over several quarters. Sentiment today has a contemporaneous effect on the interest rate, but the effect is stronger for future quarters. Positive sentiment indicating interest rate hikes in future quarters supports findings by (Shapiro and Wilson, 2021) who showed that FOMC sentiment is an indicator of target rate cuts.

Table 5: Dynamic Effect of Sentiment on Fed and ECB Policy Interest Rate

Panel A: Dynamic effect of sentiment on Federal Funds Rate					
	Fed Rate(t)	Fed Rate(t+1)	Fed Rate(t+2)	Fed Rate(t+3)	Fed Rate(t+4)
Fed(t)	2.334*** (0.596)	3.835*** (0.915)	4.953*** (1.260)	5.518*** (1.526)	6.311*** (1.762)
Fed(t-1)	1.511*** (0.511)	2.983*** (1.020)	4.134*** (1.416)	5.238*** (1.712)	5.715*** (1.880)
Fed Rate(t-1)	0.969*** (0.023)	0.884*** (0.041)	0.760*** (0.058)	0.610*** (0.070)	0.464*** (0.073)
Observations	89	89	89	89	89
R^2	0.971	0.896	0.770	0.620	0.486
Panel B: Dynamic effect of sentiment on ECB Refinancing Rates					
	ECB Rate(t)	ECB Rate(t+1)	ECB Rate(t+2)	ECB Rate(t+3)	ECB Rate(t+4)
ECB(t)	0.784*** (0.279)	1.540*** (0.465)	2.089*** (0.631)	2.526*** (0.802)	2.306** (0.990)
ECB(t-1)	0.936*** (0.218)	1.616*** (0.385)	1.986*** (0.534)	1.827*** (0.684)	1.952** (0.845)
ECB Rate(t-1)	0.989*** (0.020)	0.948*** (0.036)	0.884*** (0.049)	0.807*** (0.059)	0.719*** (0.068)
Observations	89	89	89	89	89
R^2	0.979	0.934	0.865	0.767	0.641

Note: The table reports results from regressing Federal funds (Panel A) and refinancing (Panel B) rates at $t + h$ on Fed / ECB sentiment at t and $t - 1$; we control for the lagged interest rate. We show White Standard Errors (in parentheses); * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

While central bank sentiment affects the policy rate set by the central bank, it is possible that, because of international linkages, it may also affect the rate at the other central bank. To test this hypothesis, we regress the ECB policy rate at time $t + h$ on Fed sentiment at time t . Results are reported in Panel A for Table 6. We find a large and long-lasting effect of Fed sentiment on ECB policy rates. This effect can be interpreted as reflecting the pattern that spillover effects are stronger from the US to the Eurozone (Ehrmann and Fratzscher, 2005). It also supports the notion that the Fed drives global financial conditions (Miranda-Agrippino and Rey, 2020, 2022). Panel B analyzes the other direction – the effect of ECB

sentiment on Fed interest rates. The effect, though present, is smaller and less pronounced. The effect is also less significant at longer horizons than the effect of Fed policy on ECB rates.

Table 6: Dynamic Effect of Sentiment on Policy Rates of the *other* Central Bank

Panel A: Dynamic effect of <i>Fed</i> sentiment on <i>ECB</i> Refinancing Rate					
	ECB Rate(t)	ECB Rate(t+1)	ECB Rate(t+2)	ECB Rate(t+3)	ECB Rate(t+4)
Fed(t)	0.527 (0.363)	1.508** (0.725)	2.233** (0.984)	2.588** (1.144)	3.262*** (1.263)
Fed(t-1)	1.435*** (0.464)	2.317*** (0.802)	3.031*** (1.007)	3.980*** (1.185)	4.712*** (1.284)
ECB Rate(t-1)	1.010*** (0.017)	0.991*** (0.032)	0.944*** (0.045)	0.882*** (0.050)	0.817*** (0.053)
Observations	89	89	89	89	89
R^2	0.977	0.928	0.862	0.788	0.711
Panel B: Dynamic effect of <i>ECB</i> sentiment on <i>Fed</i> Funds Rate					
	Fed Rate(t)	Fed Rate(t+1)	Fed Rate(t+2)	Fed Rate(t+3)	Fed Rate(t+4)
ECB(t)	0.905** (0.421)	1.700** (0.806)	2.205** (1.077)	2.629* (1.346)	2.632* (1.496)
ECB(t-1)	0.920** (0.438)	1.580** (0.781)	2.080* (1.101)	2.152* (1.304)	2.356 (1.479)
Fed Rate(t-1)	0.926*** (0.027)	0.810*** (0.048)	0.663*** (0.064)	0.503*** (0.074)	0.346*** (0.081)
Observations	89	89	89	89	89
R^2	0.957	0.850	0.679	0.476	0.292

Note: The table reports results from regressing the ECB refinancing rate at $t + h$ on *Fed* sentiment at t and $t - 1$ (Panel A) and the Federal funds rate on *ECB* sentiment (Panel B). We control for lagged interest rates. We report White Standard Errors (in parentheses); *p<0.1; **p<0.05; ***p<0.01.

3.2.3 Controlling for other measure of Sentiment

It is possible that Fed sentiment contains information about future policy because it reflects sentiment of the public, rather than sentiment that is specific to the monetary authority. To check if this is the case, we control for the Michigan Consumer Sentiment Index in

our analysis. The Michigan index is a survey of American households to gauge consumer expectations. This survey focuses on three pieces of information: how consumers view their personal finances, the economy as a whole in the short run, and the economy as a whole in the long run.

Table 7 reports the results of the dynamic effect of Fed sentiment when controlling for the Michigan sentiment index. We find that the Michigan index has a statistically significant effect on interest rates. However, unlike Fed sentiment the effect dies out quickly and does not build up over time. We also note that, while fed sentiment at t and $t - 1$ affects rates positively, the sign switches from t to $t - 1$ for the Michigan index. Adding the two effects together, the overall effect is close to zero. In short, our results are robust to including this alternative measure of sentiment.

Table 7: Dynmaic effect of CB and Michigan Sentiment on Fed funds rate (US only)

	Fed Rate(t)	Fed Rate($t+1$)	Fed Rate($t+2$)	Fed Rate($t+3$)	Fed Rate($t+4$)
Fed(t)	1.368*** (0.455)	2.799*** (0.948)	4.005*** (1.339)	4.829*** (1.764)	6.195*** (2.047)
Fed($t-1$)	2.284*** (0.510)	3.790*** (1.125)	4.955*** (1.531)	5.954*** (1.930)	5.940*** (2.131)
Michigan(t)	0.029*** (0.006)	0.031*** (0.011)	0.029* (0.017)	0.022 (0.023)	0.005 (0.026)
Michigan($t-1$)	-0.025*** (0.006)	-0.026** (0.012)	-0.026 (0.017)	-0.023 (0.022)	-0.007 (0.025)
Fed Rate($t-1$)	0.968*** (0.023)	0.882*** (0.043)	0.763*** (0.063)	0.621*** (0.077)	0.472*** (0.083)
Observations	89	89	89	89	89
R^2	0.976	0.903	0.776	0.624	0.486

Note: The table reports results from regressing the Federal funds rate at $t+h$ on both Fed sentiment and the Michigan sentiment index at t and $t - 1$; we control for lagged the Federal funds rate. We report White Standard Errors (in parentheses); * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

These results expand on and support the findings of (Shapiro et al., 2022), who find

that sentiment and sentiment shocks from measurements including the Michigan Consumer Sentiment are indicators of consumer sentiment and the federal funds rate. We see a similar effect with Fed and ECB sentiment, indicating that sentiment can be a strong indicator for key macroeconomic variables, including interest rates.

3.2.4 Effect of Sentiment on Exchange Rates

A key market variable that is affected by central bank policy is the exchange rate (Candian, 2021). To test if Fed or ECB sentiment affect the EUR/USD exchange rate (the price of one Euro in dollars), we analyze the dynamic effect of Fed and ECB sentiment on the FX rate using local projection analysis.

Table 8 reports the results. We find that ECB sentiment increases lead to an increase in the value of the Euro (relative to the US dollar). We also find that Fed sentiment increases lead to an increase in the value of the dollar relative to the Euro. Since we measure the exchange rate as dollars per euro, the latter effect shows up as a negative coefficient since a lower exchange rate reflects an appreciating dollar. We note that the immediate effect of central bank communications is significant for the ECB; the coefficient is positive and statistically significant. For the Fed, it is lagged sentiment that is negative (higher sentiment leads to an appreciating dollar). There is thus a bit of an asymmetry between responses. At longer horizons, the Fed effect becomes much larger and remains statistically significant, while the effect of ECB sentiment is more short-lived.

These results are consistent with our results when assessing the effects of sentiment on interest rates, where central bank sentiment significantly influences interest rates. Positive sentiment from both the Fed and the ECB is associated with higher interest rates, reflecting a tighter monetary policy stance. This relationship between sentiment and interest rates likely transmits to exchange rates, as expectations of higher interest rates can attract foreign capital, leading to currency appreciation. However, the lagged effects seen in exchange rates suggest that market participants may take time to fully incorporate central bank sentiment into their expectations, or that other factors, such as economic data releases and geopolitical

events, might affect this impact over time.

Table 8: Dynamic effect of Fed and ECB sentiment on the USD/EUR Exchange rate

	FX Rate(t)	FX Rate($t+1$)	FX Rate($t+2$)	FX Rate($t+3$)	FX Rate($t+4$)
Fed(t)	0.083 (0.079)	0.057 (0.129)	-0.032 (0.157)	-0.022 (0.159)	-0.103 (0.168)
Fed($t-1$)	-0.134** (0.063)	-0.217** (0.109)	-0.213 (0.132)	-0.309** (0.157)	-0.362** (0.160)
ECB(t)	0.087** (0.044)	0.155** (0.078)	0.124 (0.096)	0.095 (0.114)	0.151 (0.119)
ECB($t-1$)	-0.019 (0.053)	-0.068 (0.081)	-0.041 (0.101)	0.042 (0.118)	0.054 (0.117)
FX Rate($t-1$)	0.950*** (0.041)	0.867*** (0.065)	0.788*** (0.073)	0.719*** (0.071)	0.677*** (0.069)
Observations	89	89	89	89	89
R^2	0.914	0.775	0.651	0.572	0.538

Note: The table shows results from regressing quarterly average USD/EUR exchange rate (from FRED) at $t + h$ on Fed and ECB sentiment at t and $t - 1$; we control for the lagged Fed and ECB sentiment and for the lagged exchange rate. White Standard Errors (in parentheses); * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

4 Potential drivers of Sentiment

Having analyzed the dynamic effect of sentiment on important macro variables, we next ask if there are, in turn, dynamic drivers of sentiment.

4.1 Do Stock Returns Affect Sentiment?

It is likely that Central Bank sentiment reflects publicly available information about the economy. The stock market also reflects that information and updates as soon as new information becomes available. We therefore next ask if variation in stock market returns is related to Central Bank sentiment. To test this hypothesis, we explore the dynamic

relationship between sentiment and stock market return in the same and previous periods.

When analyzing the relationship between market performance and Central Bank sentiment, we modify our approach from our previous method. Instead of breaking the data up into quarters, we use the publication dates of the CB communication. This doubles our frequency of observations and allows us to see a direct effect between market returns and announcements. This change in frequency means a change in the filtering of our sentiment data. When we previously used the HP filter, we used a quarterly lambda value of 1600. For a frequency of eight times per year, we will use the findings of (Ravn and Uhlig, 2002) to calculate a new lambda value.⁵

When stock markets perform well, indicating investor confidence and potentially reflecting expectations of future economic growth, central bank communications may reflect this positive outlook. Conversely, when markets are in distress, the tone of central bank communications may become more cautious or pessimistic, mirroring these market conditions.

Table 9 reports the results of our local projection analysis. We do not find a contemporaneous effect; rather sentiment responds to market returns with a lag. This is true both for the Fed and the ECB.

For the Fed we find that stock returns in the six-week period ending at $t - 2$, so twelve weeks before the communication, has a statistically significant effect on sentiment, as do the previous periods. At longer horizons, the effect becomes smaller, as we might expect. The results indicate that central bank sentiment is slow to respond to market performance. This observation underscores the non-instantaneous nature of sentiment adjustments in central bank communications relative to market dynamics. These insights are useful for understanding the dynamics between market performances and central bank sentiment, highlighting the potential for central bank communications both reacting to market moves as well as anticipating future policy.

These findings support (Shapiro and Wilson, 2021) who finds that the Fed cares about stock market returns. This supports the need to identify what central banks weigh when measuring economic health and understand why those variables have a strong influence over

⁵ $\lambda = 1600 * (Frequency\ per\ year / 4)^4 = 1600 * (8/4)^4 = 25600$

central bank sentiment and the decisions that they make.

For the ECB sentiment has only a very small and lagged response to the European stock market return. The results suggests a weaker connection between stock returns and sentiment in Europe as compared to the US.

Table 9: Dynamic effect of Stock Market Returns on Sentiment

	Fed(t)	ECB(t)
Market Returns(t)	-0.007 (0.048)	-0.009 (0.011)
Market Returns(t-1)	0.085 (0.056)	-0.005 (0.013)
Market Returns(t-2)	0.247*** (0.056)	0.002 (0.013)
Market Returns(t-3)	0.123** (0.051)	0.028** (0.012)
Market Returns(t-4)	0.177*** (0.053)	-0.003 (0.011)
Market Returns(t-5)	0.092* (0.054)	-0.004 (0.014)
Fed(t-1)	0.409*** (0.074)	
ECB(t-1)		-0.388*** (0.061)
Observations	179	236
R^2	0.396	0.169

Note: Market returns are S&P 500 returns (US) and STOXX 600 (Eurozone), from Bloomberg. The difference between t and $t - 1$ is the length of time between two Fed or ECB meetings, so that we end up with approximately quarterly data. The market return data at t is measured from $t - 1$ to t . The table shows results from regressing Fed or ECB sentiment at $t + h$ on stock market return at times t to $t - 4$; we control for lagged sentiment. We report White Standard Errors (in parentheses); *p<0.1; **p<0.05; ***p<0.01.

4.2 Effect of Inflation Expectations on Sentiment

Stable inflation is, of course, one of the main macro variables that an inflation-targeting central bank tracks. It may be that heightened probabilities of extreme high inflation or deflation may trigger changes in central bank sentiment. We use the inflation disaster probabilities from Hilscher et al. (2024) to analyze this question. We measure the probability of high (above 4% average inflation) over the next five years or over the five year period starting in five years. We also measure the probability of deflation over these two horizons. We then use local projection analysis to investigate the effect of inflation expectations on sentiment.

Table 10 reports the results. We find no effect of extreme high inflation probabilities on sentiment for either the Fed (columns one and two) or the ECB (columns three and four). However, high probabilities of deflation are associated with lower sentiment. Probabilities of deflation both over the near term as well as in the future lead to lower sentiment, while deflation over the previous quarter has a positive effect, most likely capturing mean reversion.

Table 10: Dynamic effect of High Inflation/Deflation Expectations on Sentiment

	Short Run	Long Run	Short Run	Long Run
	Fed(t)	Fed(t)	ECB(t)	ECB(t)
Inflation(t)	0.487 (0.370)	0.461 (0.597)	-0.318 (0.466)	-0.709 (1.023)
Inflation(t-1)	-0.445 (0.426)	0.295 (0.590)	-0.260 (0.687)	-0.273 (0.980)
Deflation(t)	-2.651*** (0.623)	-2.100*** (0.629)	-1.495*** (0.431)	-0.943** (0.474)
Deflation(t-1)	1.600*** (0.598)	0.610 (0.599)	1.009** (0.449)	0.059 (0.506)
Fed(t-1)	0.301** (0.121)	0.212 (0.130)		
ECB(t-1)			0.262** (0.128)	0.237* (0.123)
Observations	45	45	45	45
R^2	0.502	0.163	0.359	0.457

Note: The table shows results from regressing Central Bank sentiment t on short run and long run high inflation and deflation probabilities at t and $t - 1$ (Hilscher et al., 2024); we control for lagged Central Bank sentiment. Following Montiel Olea and Plagborg-Møller (2021) we use White Standard Errors (in parentheses); *p<0.1; **p<0.05; ***p<0.01.

5 Possible Limitations

5.1 Data Requirements for Sentiment Analysis

Given the results in this paper, a natural next step could be to use our methods to analyze CB communications from other central banks, e.g. the Bank of England, Bank of Japan, or

national central banks in the Euroarea. However, not all central banks' communications can readily be analyzed using our methods.

A common problem when attempting to assess Central Bank communications is the availability of these publications and their structure. While most publications are available, they can be in forms that have to be consolidated and accurately broken up into texts. A common publishing structure for Central Banks is to have a separate page for every meeting or publication. This makes data consolidation difficult and time-consuming. This also means that even when data are collected, the format can be difficult to parse and can obscure sentence structure. An example of a step in the right direction are ECB speeches. The ECB has published a CSV file containing all speeches given by a Central Bank Member with clear labels of title, speaker, and location. While the text formatting is not perfect, doing this with all publications would vastly improve data accessibility.

5.2 Official Communications and Private Information

While we can perform analysis on the information published by Central Banks, we do not have access to all information or the private opinions of bank members. This lack of complete information poses a challenge to accurately predicting monetary policy decisions. Central Bank policymakers may have internal discussions and considerations that are not publicly disclosed, which can lead to gaps in our understanding of their policy stance and decision-making processes. For instance, the minutes and statements published by these institutions often summarize decisions and provide some context, but they do not reveal the full scope of the deliberations or the diverse views of the committee members. This incomplete picture of Central Bank Sentiment can lead to misinterpretations or oversights when conducting sentiment analysis or forecasting future policy actions.

The private communications and planning of formal communications between central bank officials can also play a crucial role in shaping monetary policy. Given how much attention these reports get and the effect they have on the market, there is extensive planning that goes into writing these reports. Planned messages may not accurately reflect the

market or future macroeconomic shocks, for example, because the Central Bank may be trying to downplay potentially bad news. This could cause situations where the sentiment measurement is positive, but officials may privately have a less positive view of economic conditions.

6 Conclusion

In this paper, we leverage natural language processing (NLP) methods, specifically FinBERT sentiment classification techniques, to capture the sentiment reflected in Central Bank communications. We analyze the dynamic effects that Fed and ECB communications has on important macro variables, including the Taylor rule and interest rates. Our analysis demonstrates the significant role that sentiment embedded within these communications plays in predicting future market variables and measures of economic conditions. Using local projection analysis, we find both long- and short-term impacts of sentiment, offering insights into the relationship between Central Bank communications and financial market movements. We demonstrate that modern automated analysis techniques can complement the previously potentially time-consuming work of sentiment labeling. These automated results not only support earlier findings of sentiment effects on market variables, but give us the ability to increase our sample size at lower computational costs.

Since our sentiment measure captures the effects of central bank communications on monetary policy, it may be useful for identifying monetary policy shocks. Using this measure of sentiment allows for increases the robustness in the analysis of (Romer and Romer, 1989) and (Romer and Romer, 2023). By automating the analysis of sentiment and doing it in a way that allows for the understanding of nuanced policy publications, allows for the expansion of data used in identifying shocks and increasing model accuracy.

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A Market Data

Data	Source
US Taylor Rule Quarterly	FRED
EU Taylor Rule Quarterly	Bloomberg
US Federal Funds Rate Quarterly	FRED
ECB Main Refinancing Operations Announcement Rate Quarterly	Bloomberg
SP500 Value weighted returns including dividends Quarterly	CRSP
STOXX600 Returns Gross Dividends Quarterly	Bloomberg